DG INTERNS HUB — TECHNICAL STUDY WRITEUP

WEEK 4 – SIEM & SECURITY MONITORING: INTRODUCTION TO SIEM

Prepared By:	**Gunjan**	**Department:**	
Internship:	**DG Interns Hub Cybersecurity**	**TryHackMe Room:**	
Submission Date:	**31 October 2026**	**Output File:**	

TABLE OF CONTENTS

1. Introduction & Room Objectives
2. Host-Centric vs Network-Centric Logs
3. Log Collection & Ingestion Protocols
4. Log Parsing & Schema Normalization
5. Correlation Engine & Detection Logic
6. Dashboard Visualization & Alert Triage
7. Real-World SOC Applications
8. Key Learnings & Conclusion

1. INTRODUCTION & ROOM OBJECTIVES

Modern enterprise networks generate billions of event logs daily across thousands of endpoints, servers, firewalls, and cloud assets. Inspecting log files individually on isolated endpoints is impossible during security incidents. Security Information and Event Management (SIEM) solves this challenge by serving as a central hub that aggregates, normalizes, correlates, and analyzes security event logs in real time.

This study writeup documents the core concepts covered in TryHackMe's *Introduction to SIEM* room. The primary learning objectives focus on understanding log data categories, ingestion pipelines, normalization schemas, correlation engine logic, and the SOC alert triage workflow.

2. HOST-CENTRIC VS NETWORK-CENTRIC LOGS

Log sources in an enterprise infrastructure are broadly divided into two foundational types:

- **Host-Centric Logs:** Data generated by host operating systems and applications. Examples include Windows Event Logs (e.g., Event ID 4624 successful logon, Event ID 4688 process creation), Linux Syslog (`/var/log/auth.log`), Sysmon operational logs, and endpoint application logs. These provide deep visibility into local process execution, user privileges, file modifications, and authentication.
- **Network-Centric Logs:** Telemetry captured from network perimeter and routing infrastructure. Examples include Firewall traffic logs (Palo Alto, Cisco ASA), DNS query logs, DHCP lease logs, Proxy access logs, and NIDS alerts (Snort, Suricata). Network logs reveal communication attempts, external connection destinations, data exfiltration volumes, and lateral traffic.

3. LOG COLLECTION & INGESTION PIPELINES

Log collection relies on standardized forwarding mechanisms to transport event messages from endpoints to central collectors without packet loss or message alteration:

- **Syslog Protocol (RFC 5424):** Standard network protocol using UDP/TCP port 514 to stream system messages from Linux systems, network routers, and firewalls. Messages contain facility codes (system origin) and severity levels (ranging from Emergency to Debug).
- **Agent-Based Collection:** Lightweight forwarder software (e.g., Splunk Universal Forwarder, Wazuh Agent, Winlogbeat) installed directly on target OS nodes. Agents read local event logs, encrypt data streams (via TLS over TCP), and buffer events locally if network connection drops.
- **Agentless Ingestion:** API-driven pulls (such as AWS CloudWatch APIs, Microsoft Graph API) or remote Windows Event Forwarding (WEF) over WinRM.

4. LOG PARSING & SCHEMA NORMALIZATION

Raw log entries arrive in disparate, unstructured formats. A Cisco firewall logs `src=10.0.0.5` while a Windows host records `Source_Network_Address: 10.0.0.5`. A SIEM parsing engine applies Regex-based key-value extraction rules to break raw strings into parsed data fields.

Normalization converts vendor-specific field names into a unified common schema (e.g., Common Event Format [CEF], Elastic Common Schema [ECS], or Splunk Common Information Model [CIM]). Normalization allows a SOC analyst to query `src_ip=10.0.0.5` across firewalls, routers, and Windows endpoints simultaneously in a single search.

```
# Unstructured Raw Firewall Log Example:
%ASA-6-302013: Built outbound TCP connection 20984 for outside:198.51.100.45/80 to inside:10.0.0.1
5/49152

# SIEM Normalized JSON Output Example:
{
  "event_type": "network_connection",
  "action": "allowed",
  "src_ip": "10.0.0.15",
  "src_port": 49152,
  "dest_ip": "198.51.100.45",
  "dest_port": 80,
  "vendor": "Cisco ASA"
}
```

5. CORRELATION ENGINE & DETECTION LOGIC

Raw logs alone create overwhelming noise. The **Correlation Engine** applies logical rules and temporal conditions across normalized events to detect multi-stage attack patterns:

- **Rule Logic:** Uses Boolean AND/OR statements to link events. For example: `IF (Failed_Logins > 10 from IP_X within 5 min) AND (Successful_Login from IP_X within 2 min) -> Trigger Alert: Possible Brute Force Success`.
- **Multi-Source Correlation:** Connects host and network events. For instance, linking an NIDS alert (Snort SQLi detection on Web Server) with a subsequent local Database log showing administrative privilege creation.

6. DASHBOARD VISUALIZATION & ALERT TRIAGE

SIEM dashboards translate complex queries into single-value counters, time-series area charts, and geographical maps. During alert triage, Tier-1 analysts evaluate incoming incidents:

- **True Positive (TP):** Alert correctly identifies malicious activity (e.g., actual brute-force attack or malware beaconing). Requires immediate isolation and incident escalation.
- **False Positive (FP):** Alert triggers on benign activity (e.g., legitimate admin running backup script triggering high file-read alert). Requires detection rule tuning.

7. KEY LEARNINGS & CONCLUSION

Understanding SIEM architecture provides the foundation for all Security Operations Center work. Centralized collection, parsing, normalization, and correlation allow analysts to quickly spot stealthy threats, filter out benign noise, and maintain comprehensive organizational security visibility.